Virocon Environmental Contour Analysis for ERA5 Wave Data

Python workflow for fitting a joint metocean probability model to **significant wave height** and **mean wave period**, then computing **environmental contours** for selected return periods with `virocon`.

The script reads an ERA5-style CSV file named `input.csv` with the following complete column structure:

```
datetime,swh,mwp,mwd,wind,dwi,u10,v10
```

The contour model uses:

- `swh` as significant wave height, `Hs`, in metres;
- `mwp` as mean wave period, `MWP`, in seconds;
- `mwd` as mean wave direction, required only when directional-sector analysis is enabled.

The statistical model is fitted directly using **MWP** as the period variable. The charts still use the engineering presentation convention:

```
Peak Wave Period, Tp = 1.2 × Mean Wave Period, MWP
```

Therefore, the model fit is based on `Hs-MWP`, while plots and reported peak-period values show the derived `Tp = 1.2 × MWP`.

The script generates:

- `contours.pdf` — a multi-page PDF report with one contour plot per analysis case;
- `plots/*.png` — high-resolution PNG contour figures;
- `results.txt` — a detailed text report with configuration, fitting logs, fitted-model representation and contour summaries.

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1. Purpose and scope

The script `stats_era5_data.py` performs environmental contour analysis for oceanographic time series. It is intended for engineering screening of rare combinations of wave height and wave period, including:

- omnidirectional analysis;
- directional-sector analysis based on `mwd`;
- several return periods in the same run;
- several contour-construction methods supported by `virocon`.

The principal variables are:

Variable	CSV column	Meaning	Units
`Hs`	`swh`	Significant wave height	m
`MWP`	`mwp`	Mean wave period	s
`MWD`	`mwd`	Mean wave direction	degrees

Variable	CSV column	Meaning	Units
`Tp`	derived internally	Peak wave period used for plotting/reporting only	s

The script extracts, for each contour and return period:

- maximum `Hs` along the contour;
- `MWP` associated with maximum `Hs`;
- derived `Tp = 1.2 × MWP` associated with maximum `Hs`.

The wind-related columns in the CSV are accepted but not used by the contour analysis.

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2. Input data requirements

The script expects the input file:

`input.csv`

The expected full CSV header is:

`datetime,swh,mwp,mwd,wind,dwi,u10,v10`

Only the columns required by the selected analysis are read.

2.1 Required columns

Column	Required	Used for	Units / convention
`datetime`	yes	time index	parseable date-time
`swh`	yes	significant wave height `Hs`	metres
`mwp`	yes	mean wave period `MWP`	seconds
`mwd`	only when `PERFORM_SECTOR_ANALYSIS = True`	directional-sector filtering	degrees, normally `[0, 360)`

2.2 Optional columns accepted in the CSV

Column	Used by this script	Comment
`wind`	no	may remain in the source CSV
`dwi`	no	may remain in the source CSV
`u10`	no	may remain in the source CSV
`v10`	no	may remain in the source CSV

The script reads only the required columns using `pandas.read_csv(..., usecols=...)`. Extra columns do not affect the analysis.

2.3 Example input

```
datetime,swh,mwp,mwd,wind,dwi,u10,v10
1990-01-01 00:00:00,1.42,6.10,285,7.8,300,-6.76,3.90
1990-01-01 01:00:00,1.55,6.25,288,8.1,302,-6.85,4.32
1990-01-01 02:00:00,1.37,5.95,280,7.4,297,-6.59,3.36
```

2.4 Data checks performed by the script

The script:

1. parses `datetime` as the time index;
2. checks that the required columns are present;
3. converts `swh`, `mwp` and, when needed, `mwd` to numeric values;
4. removes rows with missing key values;
5. removes rows with non-positive `Hs` or `MWP` before model fitting.

Non-positive wave heights or periods are physically invalid for the fitted probability model and are excluded.

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3. Installation

Use Python 3.11 or newer where possible.

Install the required packages:

```
pip install pandas numpy matplotlib scipy virocon
```

A clean virtual environment is recommended for reproducibility.

3.1 Windows virtual environment

```
python -m venv .venv
.venv\Scripts\activate
python -m pip install --upgrade pip
pip install pandas numpy matplotlib scipy virocon
```

3.2 Linux or macOS virtual environment

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
pip install pandas numpy matplotlib scipy virocon
```

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4. Running the script

Place `stats_era5_data.py` and `input.csv` in the same directory, then run:

```
python stats_era5_data.py
```

The script writes the outputs in the same working directory:

```
contours.pdf
results.txt
plots/
```

If `input.csv` is missing, the script creates a synthetic example `input.csv`, prints a message and exits. Replace the synthetic file with real project data and run the script again.

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5. User configuration

The script contains a configuration block near the top of the file.

5.1 File and column settings

```
INPUT_FILE = "input.csv"
TIME_COL = "datetime"
HS_CSV_COL_NAME = "swh"
MWP_CSV_COL_NAME = "mwp"
MWD_CSV_COL_NAME = "mwd"
```

These variables define the input file and the CSV columns used by the analysis.

5.2 Peak-period plotting convention

```
TP_FROM_MWP_RATIO = 1.2
TP_PLOT_COL_NAME = "_tp_from_mwp"
```

The model is fitted with `MWP`. The derived peak-period column is used only for plots and output summaries:

```
Tp = 1.2 × MWP
```

5.3 Directional-sector analysis

```
PERFORM_SECTOR_ANALYSIS = True
START_DIR_DEGREES = 0
SECTOR_WIDTH_DEGREES = 30
```

When sector analysis is enabled, the script runs:

1. one omnidirectional analysis;
2. one additional analysis for each directional sector.

With the default values, the sectors are:

```
0-30, 30-60, 60-90, ..., 330-0 degrees
```

The lower sector boundary is inclusive and the upper boundary is exclusive.

5.4 Contour parameters

```
SEA_STATE_DURATION_HOURS = 3.0
RETURN_PERIODS_YEARS = [1, 5, 10, 25, 50, 100, 250]
MIN_SAMPLES_FOR_FIT = 200
```

- `SEA_STATE_DURATION_HOURS` defines the duration of one statistically independent sea state.

- `RETURN_PERIODS_YEARS` defines the return periods to compute.
- `MIN_SAMPLES_FOR_FIT` prevents fitting a model to very small samples.

5.5 Plot and output settings

```
SWAP_AXES_CONTOUR_PLOT = True
CONTOUR_METHOD_TYPE = "IFORM"
PDF_OUTPUT_FILE = "contours.pdf"
PNG_OUTPUT_DIR = "plots"
RESULTS_TXT_FILE = "results.txt"
```

When `SWAP_AXES_CONTOUR_PLOT = True`, the plot axes are:

- x-axis: `Tp = 1.2 × MWP`;
- y-axis: `Hs`.

Supported contour methods are:

```
IFORM
ISORM
HighestDensity
DirectSampling
```

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6. Workflow implemented by the script

The script performs the following sequence:

1. loads `input.csv`;
2. validates and cleans the required columns;
3. derives the plotting-only peak-period column `Tp = 1.2 × MWP`;
4. defines the predefined `virocon` OMAE 2020 `Hs-Tz` model structure;
5. supplies `Hs` and `MWP` to the bivariate model-fitting process;
6. fits the joint probability model;
7. computes environmental contours for the selected return periods;
8. plots the data and contours using the derived `Tp` axis;
9. writes PNG plots and a multi-page PDF;
10. repeats the analysis for directional sectors, if enabled;
11. writes the detailed `results.txt` report.

The period column passed into the `virocon` model uses the internal model field name `zero_upcrossing_period`, because that is the predefined model interface. In this script, that field is populated with the CSV `mwp` values. No `pp1d` or peak-period column is required in the input file.

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7. Period convention used by the script

The input period is:

```
MWP = mean wave period
```

The fitted joint model uses:

```
Hs-MWP
```

The plotted and reported peak period is derived as:

```
Tp = 1.2 × MWP
```

This convention preserves the existing chart style while ensuring that the joint distribution is fitted with the `mwp` column now present in the CSV.

The derived `Tp` values should be interpreted as plotting and reporting aids, not as independently fitted peak-period data.

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8. Environmental contour theory

Environmental contours describe rare combinations of environmental variables. In this script the variables are `Hs` and `MWP`.

For a target return period `T_R`, the sea-state exceedance probability is computed from the sea-state duration:

```
alpha = sea_state_duration_hours / (return_period_years × 365.25 × 24)
```

The script delegates this calculation to:

```
calculate_alpha(SEA_STATE_DURATION_HOURS, rp_years)
```

The fitted joint model gives the probability structure of the wave climate. The selected contour method then constructs a curve associated with the target exceedance probability.

8.1 Joint modelling

The bivariate probability structure may be written conceptually as:

```
f(Hs, MWP) = f(Hs) × f(MWP | Hs)
```

This is more physically meaningful than treating wave height and wave period as independent, because period statistics generally depend on sea-state severity.

8.2 Predefined Virocon model

The script uses:

```
get_OMAE2020_Hs_Tz()
```

This predefined model provides a global hierarchical model structure for wave-height and wave-period contour analysis. The script uses that structure while supplying the actual period data from the `mwp` column.

8.3 Contour methods

The default method is `IFORM`.

The available methods are:

Method	General meaning
`IFORM`	Inverse First-Order Reliability Method

Method	General meaning
`ISORM`	Inverse Second-Order Reliability Method
`HighestDensity`	contour based on a highest-density probability region
`DirectSampling`	sampling-based contour construction

Different contour methods may generate different curves from the same fitted joint model. This is expected and reflects different exceedance-region definitions.

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9. Directional-sector analysis

When `PERFORM_SECTOR_ANALYSIS = True`, the script uses `mwd` to split the dataset into directional sectors.

9.1 Sector construction

The sectors start at:

```
START_DIR_DEGREES
```

and advance by:

```
SECTOR_WIDTH_DEGREES
```

until the full 360° circle is covered.

If the sector width does not divide 360 exactly, the final sector is shortened to avoid gaps or overlap.

9.2 Sector interval convention

The script uses:

```
lower bound inclusive
upper bound exclusive
```

Example:

```
30-60 means 30° <= mwd < 60°
```

9.3 Wrap-around sectors

Wrap-around sectors are handled explicitly.

Example:

```
340-10 means mwd >= 340° OR mwd < 10°
```

9.4 Direction convention

The script does not convert directional conventions. It assumes that `mwd` is already expressed in the convention intended for the engineering analysis.

Before using directional results, confirm whether the source data direction means:

- waves coming from that direction; or
- waves travelling towards that direction.

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10. Outputs

10.1 `contours.pdf`

The PDF report contains one page for each successful analysis case:

- omnidirectional case;
- each directional sector with enough data, when sector analysis is enabled.

Each plot includes:

- scatter plot of the available data;
- contours for the selected return periods;
- labels near the maximum-`Hs` points;
- a compact table with `Hs`, `MWP` and derived `Tp` at maximum `Hs`.

10.2 `plots/`

The script writes one high-resolution PNG figure per analysis case:

```
plots/contour_omnidirectional.png
plots/contour_sector_0-30_deg.png
...
```

The actual filenames are sanitized automatically from the analysis titles.

10.3 `results.txt`

The text report contains:

- overall results summary table;
- configuration used in the run;
- number of data points loaded;
- model-fitting attempt logs;
- fitted model representation;
- contour maxima for each return period;
- warnings for skipped sectors or failed contour calculations.

The summary table columns are:

```
Analysis Case
Return Period (yr)
Max Hs (m)
MWP @ Max Hs (s)
Tp @ Max Hs (s)
```

---

11. Robust model fitting and fallback logic

The script includes several robustness measures for practical metocean datasets.

11.1 Minimum sample count

Before fitting, each dataset or sector must contain at least:

```
MIN_SAMPLES_FOR_FIT = 200
```

Cases with fewer samples are skipped.

11.2 Positive-value filtering

The script removes rows where:

```
Hs <= 0
MWP <= 0
```

These values are incompatible with the fitted distributions.

11.3 Fitting strategies

The script tries the OMAE 2020 hierarchical model using:

1. default fit descriptions;
2. perturbed initial guesses;
3. widened parameter bounds.

This improves resilience when fitting difficult sectors.

11.4 Fallback model

If all full hierarchical fitting attempts fail, the script tries a simplified independent fallback model.

The fallback model is a practical recovery mechanism. It should not be interpreted as equivalent to the preferred dependent `Hs-MWP` model. If fallback results are used, inspect the affected sector or dataset carefully.

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12. Interpreting the results

The maximum `Hs` point on a contour is a useful screening value, but it is not necessarily the governing design state for every structure.

A structure or operation may instead be governed by:

- longer period at lower `Hs`;
- wave steepness;
- directional exposure;
- resonance-sensitive period ranges;
- operational or mooring response criteria;
- overtopping, armour stability or run-up mechanisms.

For engineering use, contour points should normally be checked against the relevant response model, not only against the maximum wave height.

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13. Engineering assumptions and limitations

13.1 The method is probabilistic

A 100-year contour does not mean that every point on the curve occurs once every 100 years. It is a contour associated with a target exceedance probability derived from the return period and sea-state duration.

13.2 The derived peak period is approximate

The script assumes:

```
Tp = 1.2 × MWP
```

This is an engineering convention for plotting/reporting in this workflow. It is not a replacement for measured or modelled `Tp` data.

13.3 The input direction convention must be verified

Directional-sector results depend directly on the interpretation of `mwd`. The script does not decide whether directions are "coming from" or "going to".

13.4 Sector width is a modelling choice

Narrow sectors provide better directional resolution but fewer data points. Wider sectors provide more stable fits but may mix different wave climates.

13.5 Fallback contours require caution

Fallback contours may be useful diagnostically, but they do not preserve the same dependent period-height structure as the full hierarchical model.

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14. Troubleshooting

14.1 `ModuleNotFoundError: No module named 'virocon'`

Install dependencies:

```
pip install pandas numpy matplotlib scipy virocon
```

14.2 Missing input columns

Check that `input.csv` contains the required columns:

```
datetime,swh,mwp
```

If sector analysis is enabled, it must also contain:

```
mwd
```

The script no longer requires a `pp1d` column.

14.3 The script created a synthetic `input.csv`

This means `input.csv` was not found. Replace the generated synthetic file with real data and run the script again.

14.4 Many sectors are skipped

Likely cause: fewer than `MIN_SAMPLES_FOR_FIT` rows in those sectors.

Possible adjustments:

- increase sector width;
- reduce `MIN_SAMPLES_FOR_FIT` only if justified;
- use a longer time series;
- disable sector analysis.

14.5 Fitting fails repeatedly

Possible causes include:

- insufficient data;
- strongly clustered sector data;
- extreme outliers;
- non-representative period-height dependence;
- unsuitable directional subdivision.

Suggested checks:

- inspect the raw data;
- check units;
- check whether `mwp` is positive and realistic;
- check direction convention;
- test wider sectors;
- inspect `results.txt` for the exact fitting messages.

14.6 Plots look correct but period values seem too high or low

Remember that the plotted period is not the raw `mwp`. It is:

```
Tp = 1.2 × MWP
```

The model fit and contour calculation use `MWP`.

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15. References

- Virocon documentation: https://virocon.readthedocs.io/
- Virocon GitHub repository: https://github.com/virocon-organization/virocon
- Haselsteiner, A. F., Sander, A., Ohlendorf, J. H., and Thoben, K.-D. (2020). *Global hierarchical models for wind and wave contours: physical interpretations of the dependence functions.* OMAE 2020.
- Haselsteiner, A. F., Lehmkuhl, J., Pape, T., Windmeier, K.-L., and Thoben, K.-D. (2019). *ViroCon: A software to compute multivariate extremes using the environmental contour method.* SoftwareX.
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